KATHFORD INTERNATIONAL COLLEGE OF ENGINEERING AND MANAGEMENT

DEPARTMENT OF COMPUTER AND ELECTRONICS COMMUNICATION & INFORMATION ENGINEERING

A

MINOR PROJECT REPORT

ON

“SURVEILLANCE ROVER WITH RESCUE ASSISTANCE FEATURE”

By:

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Lalitpur, Nepal

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KATHFORD INTERNATIONAL COLLEGE OF ENGINEERING AND MANAGEMENT
BALKUMARI, LALITPUR

"SURVEILLANCE ROVER WITH RESCUE ASSISTANCE FEATURE"

[Subject Code: EX654]

By:

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A PROJECT WAS SUBMITTED TO THE DEPARTMENT OF COMPUTER AND ELECTRONICS, COMMUNICATION & INFORMATION ENGINEERING IN PARTIAL FULLFILLMENT OF THE REQUIREMENT FOR THE BACHELOR'S DEGREE IN ELECTRONICS, COMMUNICATION & INFORMATION / COMPUTER ENGINEERING

DEPARTMENT OF COMPUTER AND ELECTRONICS, COMMUNICATION & INFORMATION ENGINEERING

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BALKUMARI, LALITPUR

DEPARTMENT OF COMPUTER AND ELECTRONICS, COMMUNICATION & INFORMATION ENGINEERING

The undersigned certify that they have read, and recommended to this department for acceptance, a project report entitled **"Surveillance Rover with Rescue Assistance Feature"** submitted by **Srijan Khadka**, **Sujan Shrestha** and **Shankar Rai** in partial fulfilment of the requirements for the Bachelor's degree in Electronics, Communication & Information Engineering.

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DATE OF APPROVAL: 6th MARCH 2024

DEPARTMENTAL ACCEPTANCE

The project entitled "**Surveillance Rover with Rescue Assistance Feature**", submitted by **Srijan Khadka**, **Sujan Shrestha** and **Shankar Rai** in partial fulfilment of the requirements for the Bachelor's degree in Electronics, Communication & Information / Computer Engineering has been accepted as a bonafide record of work carried out by them in this department.

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In conclusion, we extend our heartfelt thanks to all individuals and entities who have played a role in the successful completion of the Surveillance Rover with Rescue Assistance Feature project. Your contributions have been invaluable, and we are deeply grateful for your support.

Thank you.

ABSTRACT

Surveillance and rescue rovers are becoming crucial for handling unsafe areas where humans cannot or should not go. This project develops a special robot that can perform various tasks such as observing, inspecting the surroundings, and assisting in rescue operations. The rover has DHT11 sensor for temperature and humidity monitoring. Additionally, the rover has a Pi camera that shows a live video. This helps emergency workers get a clear picture of the situation during rescue missions. Users can conveniently monitor this data through web page. The project shows how we take care of the environment, handle emergencies more effectively, and use live video for monitoring and communication. The rover's compact size allows it to access hard-to-reach areas, reducing monitoring costs and overall development expenses. The Surveillance Rover with Rescue Assistance Feature project represents a significant advancement in remote monitoring and rescue technology. Through the integration of Raspberry Pi, Arduino, sensors, and a Pi Camera, the project successfully delivers a functional surveillance rover capable of real-time video streaming and environmental monitoring.

Keywords: Arduino Uno, DHT11, Raspberry Pi 4B, Surveillance Rover

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LIST OF ABBREVIATIONS

CCTV	:	Closed Circuit Television
GUI	:	Graphical User Interface
I2C	:	Inter-Integrated Communication
LiDAR	:	Light Detection and Ranging
NASA	:	National Aeronautics and Space Administration
ROVER	:	Remotely Operated Video Enhanced Receiver
Wi-Fi	:	Wireless Fidelity

1. INTRODUCTION

1.1 Background

Surveillance and rescue rovers are advanced robotic systems designed to operate in hazardous environments, providing real-time data and assistance during surveillance missions and rescue operations. These robotic platforms have gained significant attention due to their potential to revolutionize various industries, particularly in scenarios where human intervention is limited or unsafe. The 21st century has witnessed a surge in natural disasters, industrial accidents, and complex security threats. Such incidents demand effective surveillance and rescue solutions that can rapidly assess the situation, provide critical data, and assist in locating and rescuing individuals in distress. Traditional surveillance methods often fall short in hazardous or inaccessible environments, underscoring the necessity for robust robotic systems like surveillance and rescue rovers.

The development of surveillance and rescue rovers drives technological innovation, leading to the creation of more advanced robotics, sensors, and communication systems, with potential applications in various other fields. Surveillance rovers play a significant role in exploring uncharted or extreme environments, such as space, polar regions, or deep-sea areas. They can collect valuable data, which can contribute to scientific research, resource exploration, and understanding our planet better. Using surveillance and rescue rovers in hazardous environments reduces the risk to human rescuers. These machines can be sent into dangerous situations, such as collapsed buildings or toxic environments, without endangering human lives. Rovers can be used for monitoring and collecting data on environmental parameters, such as temperature and humidity, and wildlife habitats, aiding in environmental protection and conservation efforts. Overall, the development of surveillance and rescue rovers significantly contributes to public safety, disaster response, exploration, and technological progress, making them indispensable tools in today's world.

1.2 Problem Statement

In real-field scenarios, existing surveillance and rescue rovers face significant challenges in effectively conducting surveillance missions and rescue operations. The current system of surveillance and rescue rovers exhibits limitations in terms of mobility, sensing capabilities, and autonomy, hindering their ability to operate optimally in hazardous environments.

1. Conventional monitoring is outdated, unsuitable for large areas, hazardous environments, and disease-prone spaces, posing risks to humans and animals.
2. Traditional disaster data collection lacks real-time information, leading to delays in decision-making and coordination.
3. CCTV has blind spots, requires numerous cameras, and incurs high installation and maintenance costs.

Addressing these problem statements is essential for the advancement of surveillance, monitoring, and rescue systems. By overcoming these challenges, the efficiency, reliability, and effectiveness of these systems can be significantly enhanced, enabling better responses to critical missions, and contributing to safer and more resilient operations.

1.3 Objectives

The aim of this project is to develop an efficient, reliable, and effective rover system that can significantly enable better responses to critical missions and contributing to safer and more resilient operations.

1.3.1 Main Objective

i. The main objective of this project is to develop a versatile rover for surveillance and environmental monitoring with rescue assistance feature.

1.3.2 Specific Objective

i. To develop a rover capable of moving around controlled through RF medium.
ii. To monitor and record the temperature and humidity at the surveillance location.
iii. To capture the video in real time.

1.4 Scope

The project is basically an integration of hardware and software to conduct a comprehensive analysis of surveillance needs and environmental conditions to determine the rover's specifications and capabilities.

- Design and develop a remote-controlled rover equipped with cameras and sensors.
- Enable real-time monitoring of designated areas for security and surveillance purposes.

2. LITERATURE REVIEW

In the past few years, people have been very interested in using robots to keep an eye on and assist in dangerous situations. This is because it can make things more secure for people and complete their tasks more quickly. At the beginning, scientists were mostly focused on creating robots that could go to difficult places. [1]. As technology improved, researchers used more advanced tools like LiDAR, cameras, communication systems, and thermal imaging to improve their ability to monitor and gather information [2]. The development of artificial intelligence and machine learning has been very useful in making decisions on their own and analyzing data quickly. The story of self-driving surveillance and rescue vehicles began in the early 1900s. During that time, simple vehicles were operated from a distance for military activities. Over time, the initial models have developed into more sophisticated systems that are essential for investigating space, such as NASA's Mars rover [3]. Following the events of 9/11 and other natural disasters, the use of self-governing surveillance and rescue vehicles became valuable in managing these emergencies. These vehicles were used to get to places that were difficult to reach. considered too dangerous for people to take part in [4].

Current systems like CCTV monitoring and mobility have a lot of problems. Scientists have found many ways to solve a problem, and one of those ways is by using a vehicle called a rover or a self-driving vehicle. The rovers that are currently available for buying are either too expensive and complicated to operate or not efficient enough for accurate tracking. And you can't use them in small and tight places because they are too large [5]. The proposed system or robot is small and works very effectively in small areas [6]. The rover is using a tool that tells it its current location and the current time. The GPS module is a device that helps locate where something is and make sure its clock is accurate [7]. If the time on the device is different from the time on the GPS by more than 3 seconds, the device's time will be changed to match the GPS time. A lot of self-driving rovers use solar panels to make electricity. Over time, if dust accumulates or if solar panels get broken, they do not function properly, which can impact the amount of power autonomous rovers have [8]. Additionally, the battery that comes with the device can deteriorate over time, resulting in a shorter lifespan and reduced ability to hold a charge. And it costs a lot of money to make a rover like this. So, to solve the problems, the rover has batteries that can be charged rather than using solar panels. This is a less expensive choice for our project [9].

This project is about making a little device that uses IoT technology to keep an eye on small and cramped areas. It will be cheap and have sensors that collect information and have navigation and cameras for keeping track of things. IoT technology is a system where objects can talk to each other using computer addresses. This makes gadgets and items stronger [10].

The idea is to connect all the sensors and parts of the rover to the Internet, so that people can easily and instantly check the environment from a different place. IoT technology helps us make easy-to-use apps for computers or smartphones. These apps can tell the rover what to do and display information from its sensors and camera [11].

3. REQUIREMENT ANALYSIS

3.1 Feasibility Study

A feasibility study on surveillance, monitoring, and rescue rovers would typically involve assessing the technical, economical, operational, and legal aspects of the project. Here are some key components that would be considered in such a study:

3.1.1 Technical Feasibility

All the technical resources needed for the project, such as hardware components and software, are readily accessible on the market. Obtaining any additional items necessary for the project is expected to be trouble-free. So, the project is technically feasible.

3.1.2 Economical Feasibility

The project we are undertaking is financially viable and falls within a manageable budget since a significant portion of the required equipment and electronic devices are readily accessible. Once the development of the rover system is completed and it performs as intended, all the expenses incurred during the project will be justified, as we will have a comprehensive system capable of carrying out surveillance, monitoring, and rescue operations.

3.1.3 Operational Feasibility

Ensuring the effective functioning of the rover in diverse conditions requires thorough assessments of its reliability, durability, and maintenance requirements. Integrating the rover into existing emergency response systems necessitates a feasibility study to ensure seamless compatibility and optimization of its capabilities within the infrastructure. Equally important is understanding the training needs and skill requirements of emergency responders to operate and maintain the rover proficiently. By addressing these aspects comprehensively, we can establish a robust emergency response system, leveraging the rover's potential to safeguard lives and property during crises efficiently.

3.2 Hardware Requirements

At its core, the hardware setup comprises of:

- Raspberry Pi
- Pi Camera
- Arduino uno
- DHT11
- DC Geared Motor
- RF- Transmitter Receiver
- Electronic Speed Controller
- Battery
- Buck Converter

Description of above components are as follows:

3.2.1 Raspberry Pi

The Raspberry Pi 4 Model B is a highly versatile single-board computer developed by the Raspberry Pi Foundation, boasting significant improvements in processing power, connectivity, and multimedia capabilities compared to its predecessors. Powered by a quad-core ARM Cortex-A72 processor clocked at up to 1.5 GHz, the Raspberry Pi 4 offers multiple RAM options ranging from 2GB to 8GB, enabling smoother multitasking and enhanced performance in various applications. Its comprehensive connectivity features include dual-band 2.4 GHz and 5 GHz Wi-Fi, Gigabit Ethernet, Bluetooth 5.0, and USB 3.0 ports, facilitating fast data transfer rates and seamless integration with peripherals. Raspberry Pi is used as master in our project. With its GPIO pins, users can interface with external devices and sensors, fostering innovation in electronics projects and prototyping. Supported by a variety of operating systems, including Raspberry Pi OS and Ubuntu, the Raspberry Pi 4 serves as an affordable and powerful computing platform suitable for education, hobbyist projects, and industrial applications alike.

Figure 3.1: Raspberry Pi 4B

3.2.2 Pi Camera

The Pi Camera is a compact camera module designed for use with Raspberry Pi single-board computers. It offers high-definition image and video capture capabilities, with resolutions of up to 12 megapixels for still images and 1080p for video. The Pi Camera attaches directly to the Raspberry Pi's camera port via a ribbon cable, enabling easy integration into various projects. With adjustable focus lenses and compact form factor, it's ideal for applications such as surveillance, robotics, and multimedia projects. Its affordability and compatibility with Raspberry Pi make it a popular choice for hobbyists and professionals alike.

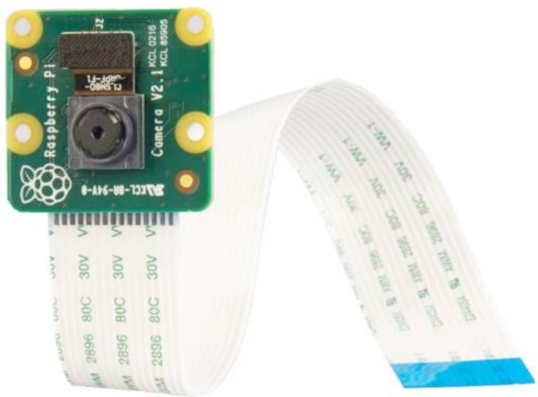


Figure 3.2: Pi Camera

3.2.3 Arduino Uno

The Arduino Uno is a versatile microcontroller board widely used for prototyping and DIY projects. It features 14 digital input/output pins, 6 analog inputs, and a 16 MHz quartz crystal oscillator. With USB connectivity, it allows easy programming and communication with computers. The Uno supports a variety of sensors, actuators, and shields, making it suitable for a range of applications from robotics to home automation. Its beginner-friendly design and extensive community support make it an ideal choice for electronics enthusiasts and students. Arduino uno is used as a slave in our project.

Table 3.1: Pin Configuration of Arduino Uno

Pin Category	Pin Name	Details
Power	Vin, 3.3V, 5V, GND	Vin: Input voltage to Arduino when using an external power source.
		5V: Regulated power supply used to power microcontroller and other components on the board.
		3.3V: 3.3V supply generated by on-board voltage regulator. Maximum current draw is 50mA.
		GND: ground pins.
Reset	Reset	Resets the microcontroller.
Analog Pins	A0 – A5	Used to provide analog input in the range of 0-5V
Input/Output Pins	Digital Pins 0 - 13	Can be used as input or output pins.
Serial	0(Rx), 1(Tx)	Used to receive and transmit TTL serial data.
External Interrupts	2, 3	To trigger an interrupt.
PWM	3, 5, 6, 9, 11	Provides 8-bit PWM output.
SPI	10 (SS), 11 (MOSI), 12 (MISO) and 13 (SCK)	Used for SPI communication.
Inbuilt LED	13	To turn on the inbuilt LED.
TWI	A4 (SDA), A5 (SCA)	Used for TWI communication.
AREF	AREF	To provide reference voltage for input voltage.

Figure 3.3: Arduino Uno

3.2.4 DHT11

The DHT11 is a basic digital temperature and humidity sensor module. It consists of a calibrated digital signal output, with the temperature and humidity sensing elements integrated within a single component. The sensor is easy to use, low-cost, and provides accurate measurements for both temperature and humidity levels in various environments. It is used for environmental monitoring and control.

Table 3.2: Pin Configuration of DHT11

S.N.	Pin Name	Description
1	Vcc	Power supply 3.5V to 5.5V
2	Data	Outputs both Temperature and Humidity through serial
3	NC	No Connection and hence not used
4	Ground	Connected to the ground of the circuit

Figure 3.4: DHT11 Sensor

3.2.5 DC Geared Motor

A DC geared motor is a type of electric motor equipped with a gearbox to increase torque and reduce speed. It operates on direct current (DC) and is commonly used in robotics, automation, and vehicles where precise control of speed and direction is required. In this project we have used 6 motors of 200 rpm which runs on 12V power supply.

Figure 3.5: DC Geared Motor

3.2.6 RF- Transmitter Receiver

RF (Radio Frequency) transmitter-receiver modules enable wireless communication between devices over short to medium distances. They transmit data using radio waves and are commonly used in remote control systems, telemetry, and wireless sensor networks.

Table 3.3: Specification of Tx-Rx

Rf range	2.4055 - 2.475 GHz
Channels	6
Bandwidth	500KHz
System with 2.4G	AFHDS 2A & AFHDS 2B
Low voltage warning	Below 4.2V
Output	PPM, SBUS, PWM
ANT length	26mm*2 (dual antenna)

Figure 3.6: RF-Transmitter Receiver

3.2.7 Electronic Speed Controller

The PowerMax 20-D is a dual channel bidirectional ESC capable of supplying 20 Amps per channel.

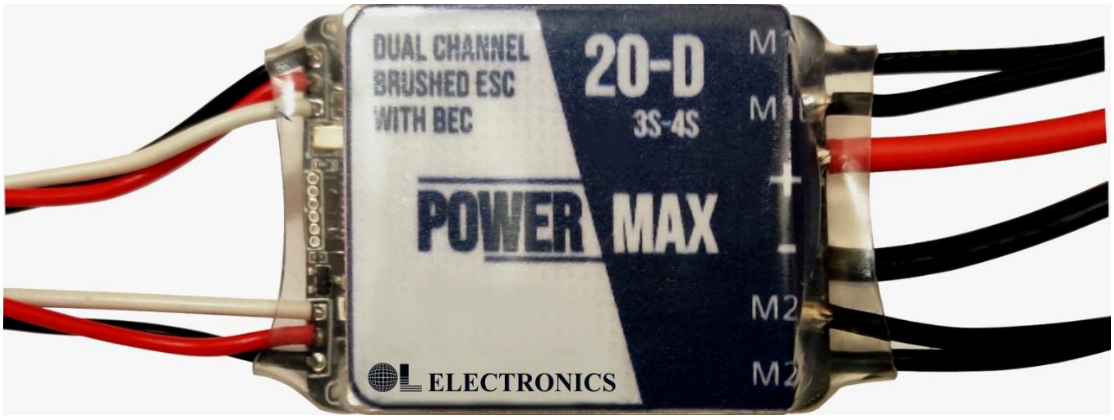


Figure 3.7: Electronic Speed Controller

3.2.8 Battery

Battery serves as a portable power source. It provides the electrical energy necessary to drive motors, power microcontrollers, and operate various electronic components. Here we have used a 12V LiPo Battery to power the electronic system.

Figure 3.8: Battery

3.2.9 Buck Converter

A buck converter or step-down converter is a DC-to-DC converter which decreases voltage, while increasing current, from its input (supply) to its output (load). It is a class of switched-mode power supply. Switching converters (such as buck converters) provide much greater power efficiency as DC-to-DC converters than linear regulators, which are simpler circuits that dissipate power as heat, but do not step up output current.[1] The efficiency of buck converters can be very high, often over 90%, making them useful for tasks such as converting a computer's main supply voltage, which is usually 12 V, down to lower voltages needed by USB, DRAM and the CPU, which are usually 5, 3.3 or 1.8 V. We have used a buck converter to supply 5V to the Raspberry Pi.

Figure 3.9: Buck Converter

3.3 Software Requirements

We have used several software for the fulfillment of the objectives of our project.

- Proteus
- Arduino IDE
- Raspbian
- SolidWorks
- HTML
- CSS
- Python

Description of above software are as follows:

3.3.1 Proteus

Proteus is a simulation software widely used for designing and simulating electronic circuits and microcontroller-based projects. In our project, Proteus can be utilized to design and simulate the electronic circuitry, including connections, components, and microcontroller behavior, before implementing it physically. This allows for testing and debugging of circuits without the need for physical hardware, saving time and resources.

3.3.2 Arduino IDE

The Arduino Integrated Development Environment (IDE) is a software application used for writing, compiling, and uploading code to Arduino microcontroller boards. In our project, the Arduino IDE is essential for writing and uploading the firmware code that controls the behavior of the Arduino microcontroller, including sensor readings, motor control, and communication protocols.

3.3.3 Raspbian

Raspbian is the official operating system based on Debian Linux designed for Raspberry Pi single-board computers. In our project, Raspbian serves as the operating system for the Raspberry Pi, providing the necessary software environment for hosting web servers, running Python scripts, and interfacing with peripherals and sensors.

3.3.4 Solidworks

SolidWorks is computer-aided design (CAD) software used for creating 3D models and assemblies. In our project, SolidWorks can be employed for designing and modeling mechanical components of the surveillance rover, such as chassis, mounts, and enclosures. It allows for precise visualization and analysis of the physical components before manufacturing.

3.3.5 HTML

HTML (Hypertext Markup Language) is the standard markup language used for creating web pages and web applications. In our project, HTML can be utilized for designing the structure and content of the web interface used to control and monitor the surveillance rover. It defines the layout, text, images, and other elements displayed on the webpage.

3.3.6 CSS

CSS (Cascading Style Sheets) is a style sheet language used for defining the presentation and layout of HTML documents. In our project, CSS can be applied to enhance the appearance and styling of the web interface, including colors, fonts, borders, and layout positioning. It ensures consistency and aesthetic appeal across different web pages and devices.

3.3.7 Python

Python is a versatile, high-level programming language known for its simplicity and readability. In our project, Python is used for controlling the Raspberry Pi, interacting with sensors, and hosting the web server for streaming video and displaying sensor data. Its extensive standard library and rich ecosystem make it suitable for a wide range of applications, from web development to data analysis and beyond. Python's ease of use and flexibility make it a popular choice for projects of all sizes and complexities.

4. METHODOLOGY

The process of making the rover involves four steps. The first step is concept design, where different studies are performed to find problems in traditional surveillance systems and the most advanced methods in environmental monitoring. The main goal is to make a small rover that can go to difficult places and is not expensive. It will check and monitor the environment. The next step involves choosing and creating the physical equipment needed, using the requirements from the initial step. During the third phase, the team creates software for the rover. This includes writing instructions for the hardware and designing a web app to keep track of environmental information. At last, during the testing stage, the rover is sent out to collect details about the area it is in.

4.1 Development Model

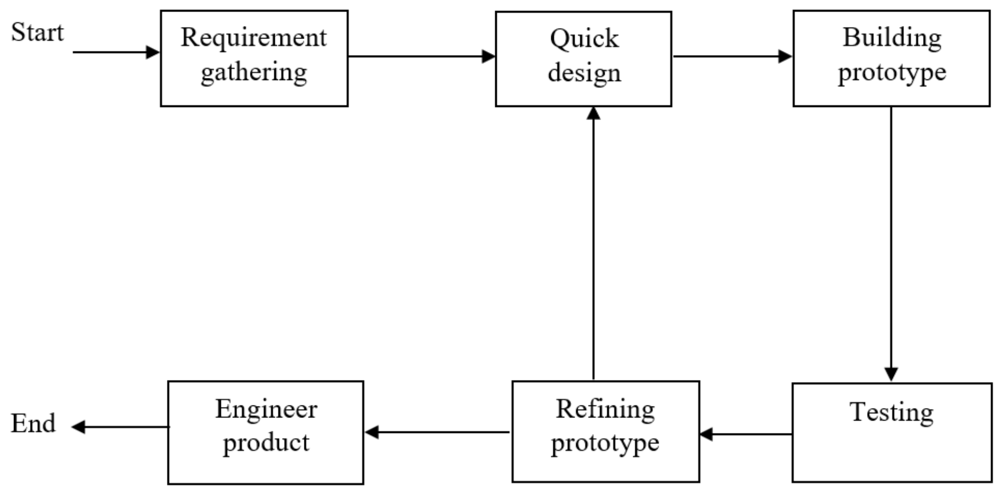


Figure 4.1: Prototyping Model

The prototyping model is a good choice for projects because it allows for iterative development, early feedback, and reduces the risk involved. When users are regularly involved, they are more likely to work together, which helps make the focus of the approach all about the user. Doing things quickly and saving money happen when you find and fix problems early. Prototyping is a popular method in project development because it is flexible and efficient.

4.2 Block diagram

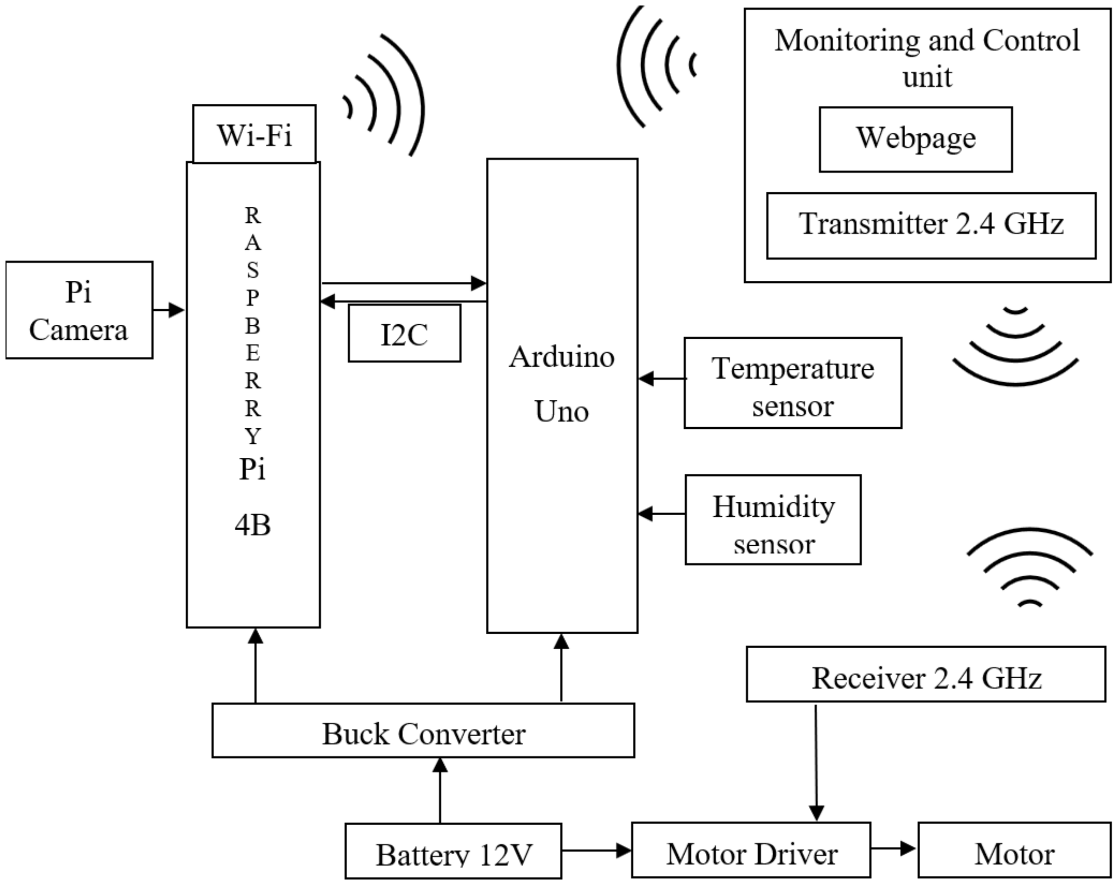


Figure 4.2: Block Diagram of Rover

The project block diagram basically shows the use of Raspberry Pi 4B as master and Arduino uno as slave. Raspberry Pi 4B has its built in Wi-Fi module. Pi camera is integrated to the Raspberry Pi while temperature and humidity sensor is integrated to the Arduino uno. The communication between Raspberry Pi and Arduino uno is I2C Communication.12 Power is supplied to motor through motor driver and to Arduino and Raspberry Pi through Buck Converter which lowers the 12V power supply to 5V. The Rover is controlled by RF Transmitter. Monitoring of sensor data and video surveillance is done in the webpage.

4.3 Flowchart

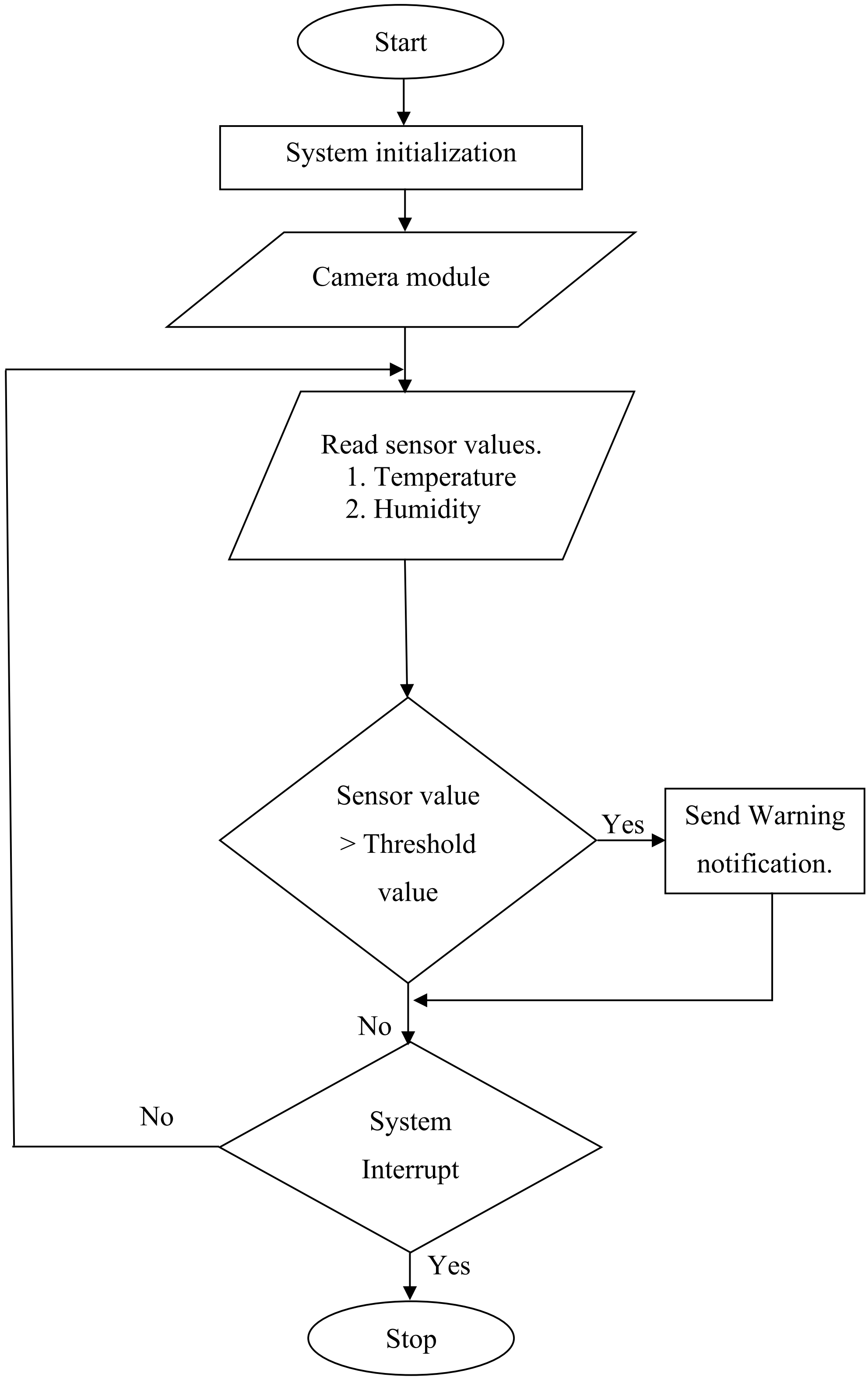


Figure 4.3: Flowchart of Rover system

5. SYSTEM DESIGN

The Surveillance Rover project integrates a range of hardware and software components to enable remote surveillance and rescue operations.

5.1 Rover Design

Firstly, the rover was designed in SolidWorks with the probable dimensions in the initial phase.

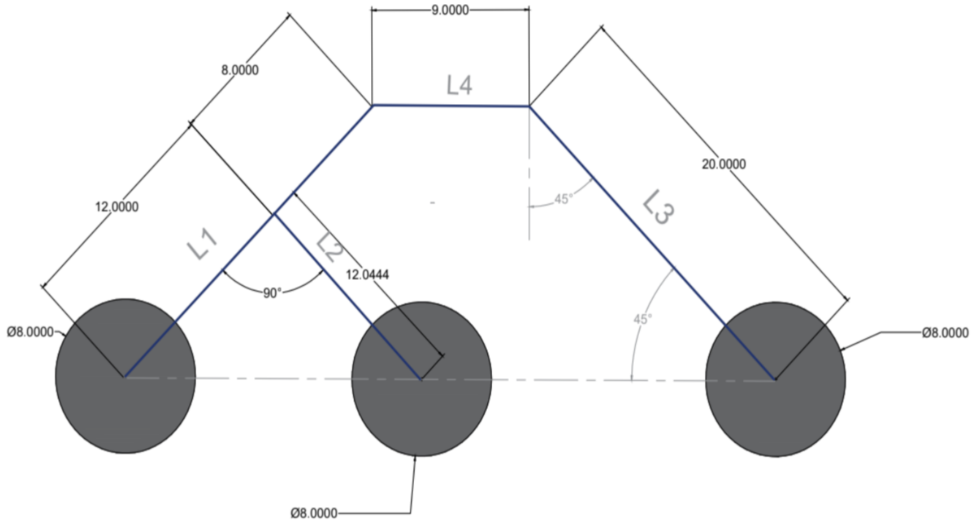


Figure 5.1: SolidWorks Rover Dimension

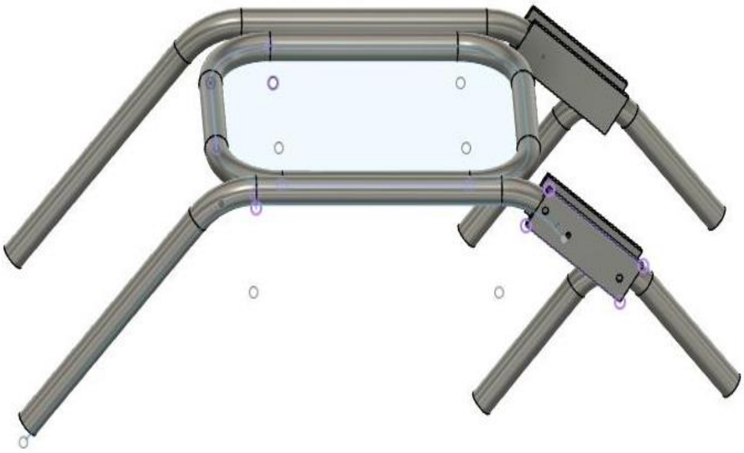


Figure 5.2: SolidWorks Rover chassis design

Finally, the SolidWorks design was converted to the original form capable of proper mobility. The rover chassis is made using CPVC pipe and it is a 6 wheeled rover. The dimension of the rover is 60cm X 40 cm X 30 cm.

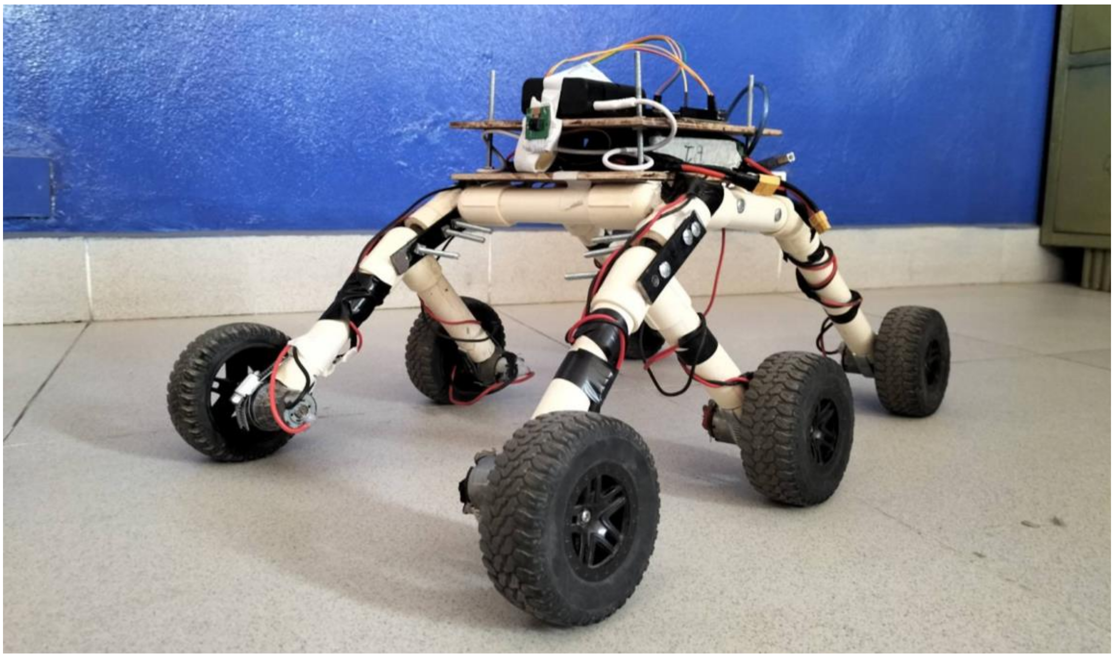


Figure 5.3: Final Rover design

5.2 Proteus Schematic Design

The project was first tested by designing the proteus schematic diagram using libraries available in the proteus before implementing it in the original components.

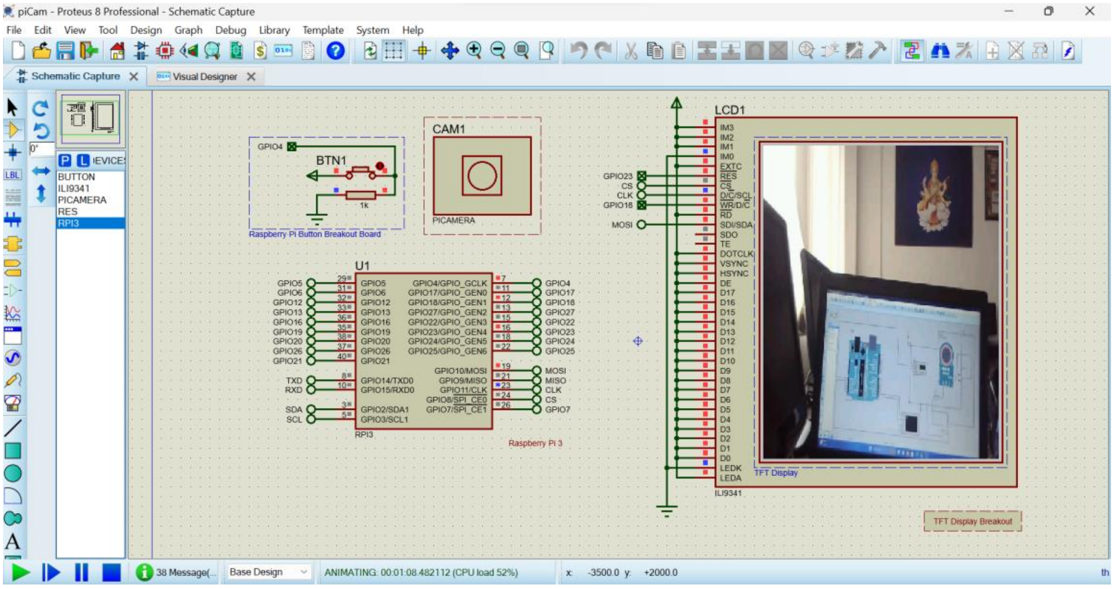


Figure 5.4: Pi Camera interface to Raspberry Pi

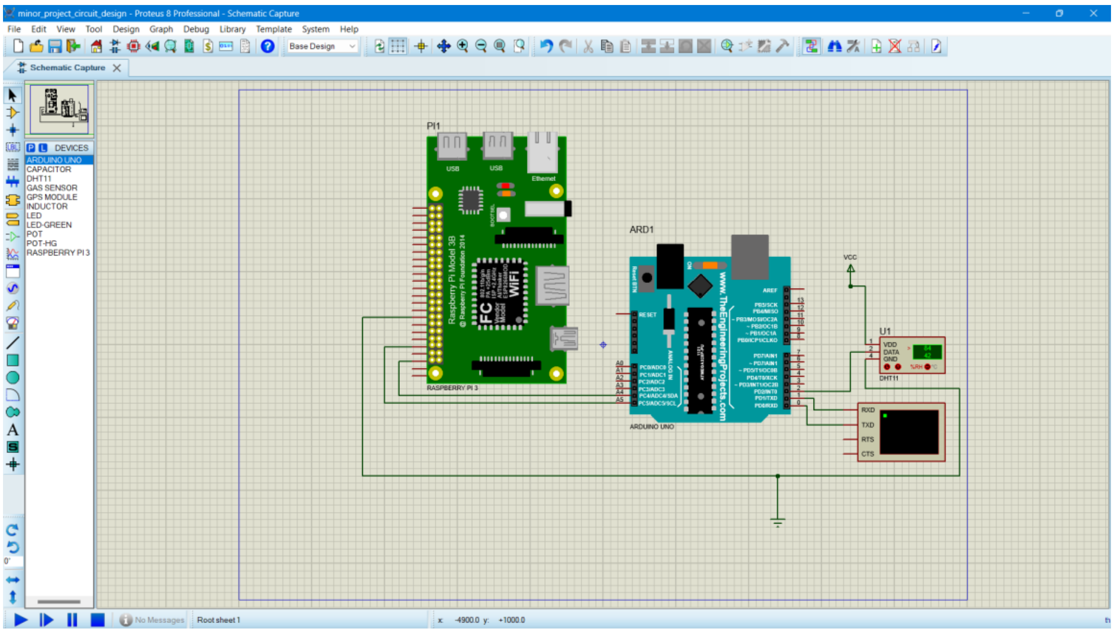


Figure 5.5: Raspberry Pi, Arduino and DHT11 interface

6. RESULT AND DISCUSSION

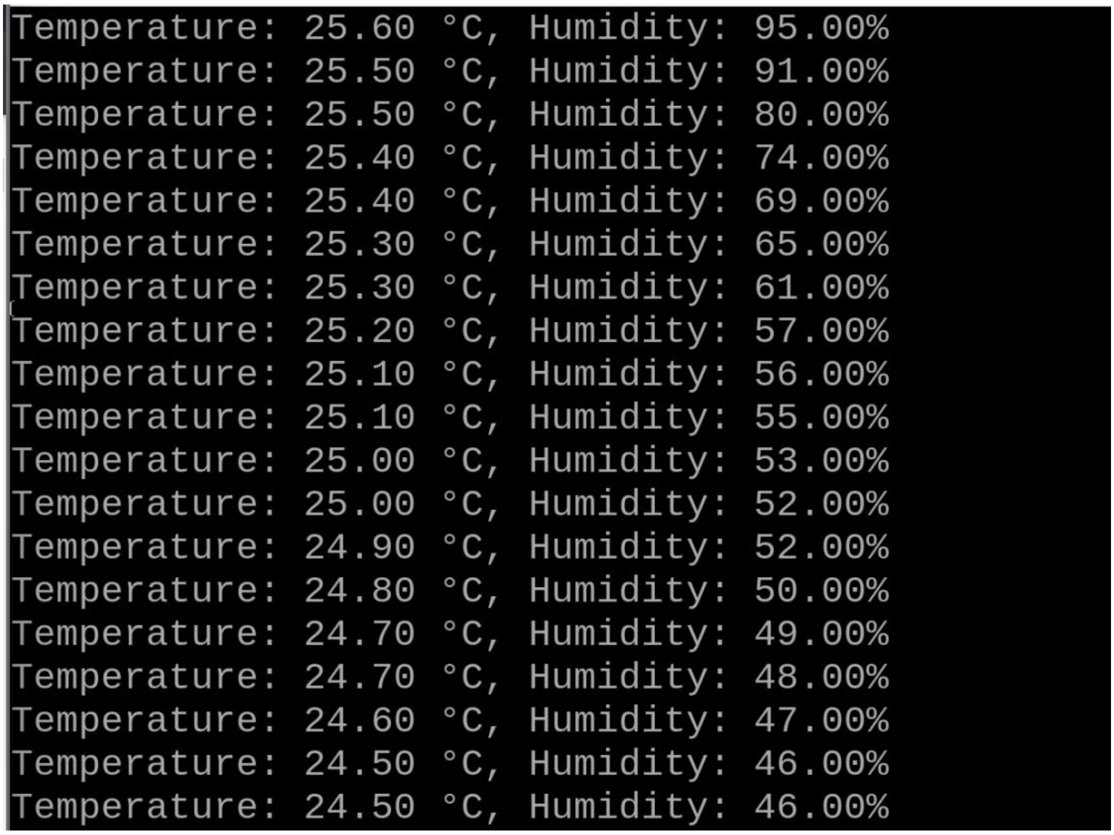
The implementation of the Surveillance Rover project yields several noteworthy results and prompts insightful discussions regarding its functionality, applications, and potential improvements. The result of our project includes:

6.1 Mobility of Rover

The rover's wireless communication system enables seamless control and data transmission between the rover and the control station. This feature enhances flexibility and accessibility, allowing operators to maneuver the rover from a distance.

6.2 Monitoring of Sensor Data

The integration of sensor DHT11 enables the rover to collect environmental data including temperature and humidity. Real-time data processing and visualization empower users to make informed decisions based on the gathered information.



```
Temperature: 25.60 °C, Humidity: 95.00%
Temperature: 25.50 °C, Humidity: 91.00%
Temperature: 25.50 °C, Humidity: 80.00%
Temperature: 25.40 °C, Humidity: 74.00%
Temperature: 25.40 °C, Humidity: 69.00%
Temperature: 25.30 °C, Humidity: 65.00%
Temperature: 25.30 °C, Humidity: 61.00%
Temperature: 25.20 °C, Humidity: 57.00%
Temperature: 25.10 °C, Humidity: 56.00%
Temperature: 25.10 °C, Humidity: 55.00%
Temperature: 25.00 °C, Humidity: 53.00%
Temperature: 25.00 °C, Humidity: 52.00%
Temperature: 24.90 °C, Humidity: 52.00%
Temperature: 24.80 °C, Humidity: 50.00%
Temperature: 24.70 °C, Humidity: 49.00%
Temperature: 24.70 °C, Humidity: 48.00%
Temperature: 24.60 °C, Humidity: 47.00%
Temperature: 24.50 °C, Humidity: 46.00%
Temperature: 24.50 °C, Humidity: 46.00%
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Figure 6.1: Sensor Data on Terminal

For testing the rover capability to monitor the environment, a test was performed at the college to check the temperature data of a day on 5th March 2024 and the result so obtained is plotted on graph below.

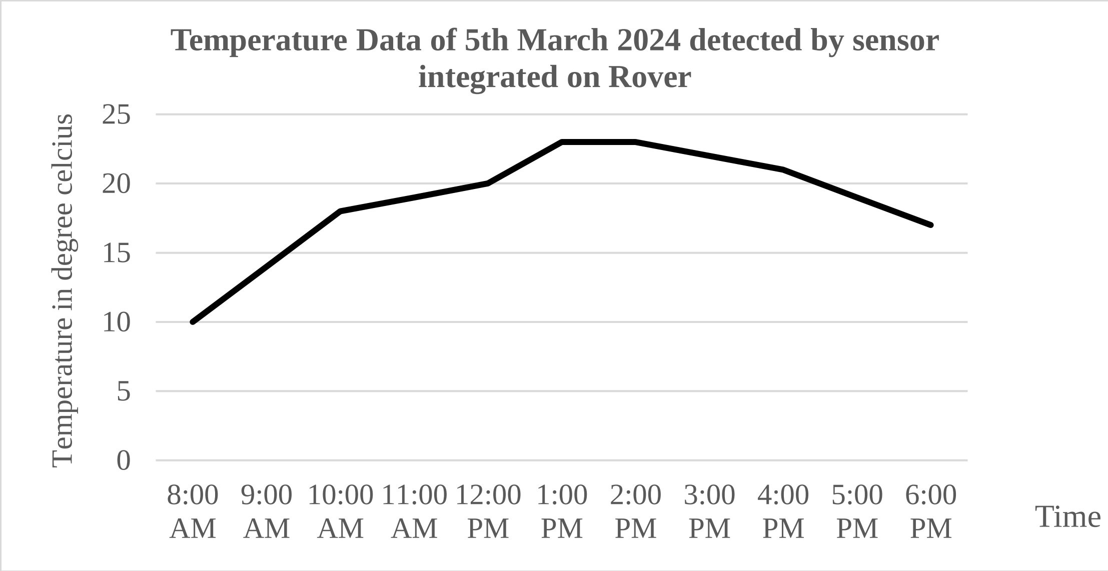

Figure 6.2: Temperature sensor data

Furthermore, for testing the data from humidity sensor, we tested at 3 different locations. Firstly, we tested inside the working room and then on the basketball court of our college and lastly in the underground. Different data was obtained which is plotted in the graph shown below.

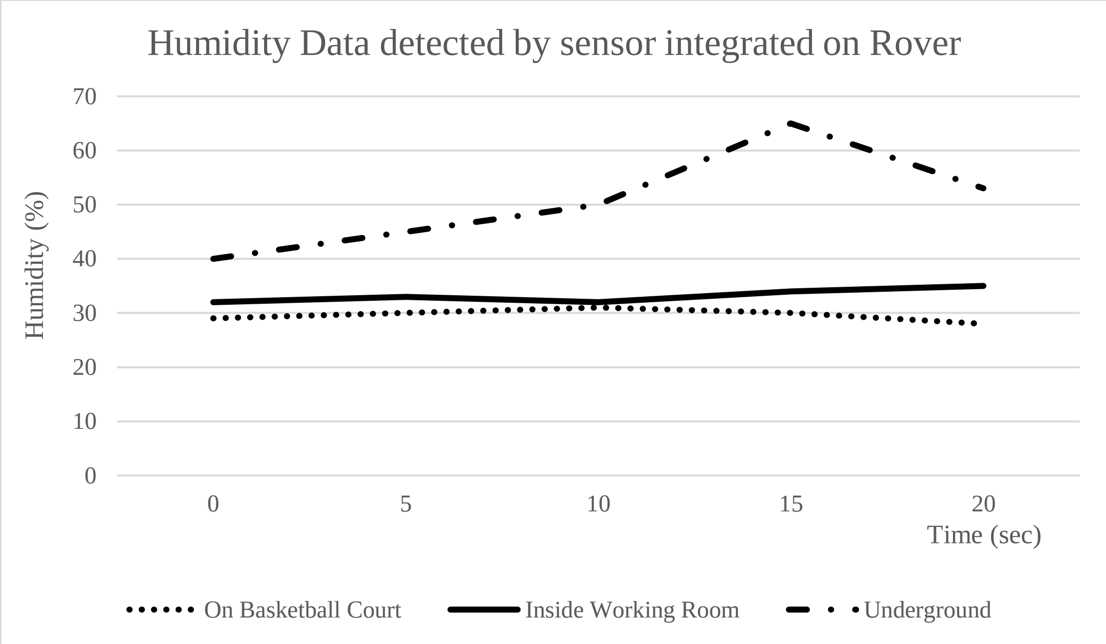

Figure 6.3: Humidity sensor data

6.3 Video Surveillance

The rover successfully provides remote surveillance capabilities, allowing users to monitor environments and access live video feeds through a web interface. This feature enhances situational awareness and facilitates monitoring of remote or hazardous locations without physical presence.

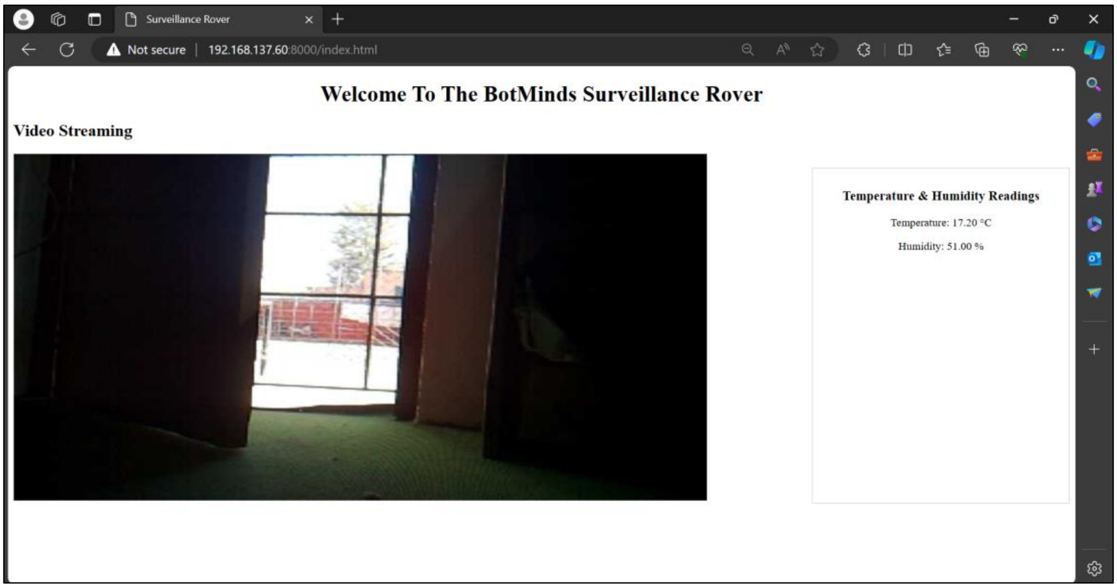

Figure 6.4: Pi Camera Surveillance

The implementation of the Surveillance Rover project prompts several discussions regarding its functionality, applications, and potential improvements. One key discussion revolves around the rover's applications in surveillance and search & rescue operations. The rover's capability to navigate different terrains and gather real-time data makes it valuable for assessing disaster areas, locating survivors, and monitoring hazardous environments. However, to maximize its effectiveness in these scenarios, enhancements in autonomous navigation algorithms are crucial. By integrating advanced algorithms, the rover could navigate complex environments more effectively and make informed decisions based on sensor data, thereby improving its utility in critical situations.

The user-friendly interface provided by the web-based platform enhances the rover's accessibility and usability. However, ongoing refinement is necessary to improve the interface's intuitiveness and efficiency. Moreover, discussions about power management and energy efficiency are crucial for optimizing the rover's performance and extending its operational lifespan in the field. By addressing these considerations, the Surveillance and Rescue Rover project can evolve into a valuable tool for enhancing situational awareness, disaster response, and search and rescue operations. Through collaborative efforts and continuous refinement, the rover can fulfill its potential as a versatile and reliable asset in critical environments.

7. CONCLUSION

In conclusion, the Surveillance Rover with Rescue Assistance Feature project represents a significant advancement in remote monitoring and rescue technology. Through the integration of Raspberry Pi, Arduino, sensors, and a Pi Camera, the project successfully delivers a functional surveillance rover capable of real-time video streaming and environmental monitoring. The project's success underscores the potential of surveillance technology in enhancing situational awareness and aiding in rescue operations during emergencies.

The project's outcomes pave the way for further advancements in surveillance and rescue assistance technology, with implications for a wide range of applications, including disaster response, security, and environmental monitoring. However, the project also highlights the importance of addressing ethical and privacy considerations associated with the deployment of surveillance technology, emphasizing the need for responsible and ethical use of such systems.

As the project concludes, reflections on lessons learned, challenges encountered, and opportunities for future research and development emerge. Continuous refinement and innovation are essential to further optimize the surveillance rover's capabilities, improve its reliability and robustness, and maximize its impact in real-world scenarios. Ultimately, the Surveillance Rover with Rescue Assistance Feature project represents a significant step forward in leveraging technology to enhance safety, security, and resilience in communities worldwide.

8. LIMITATION AND FUTURE WORK

Despite the completion of the project, still there exists some limitations which are highlighted below:

Hardware Constraints: The project's hardware components, including Raspberry Pi and Arduino, may have limitations in processing power, memory, and connectivity, which could affect the rover's performance and scalability.

Battery Life: The rover's reliance on a battery power source may limit its operational duration, especially during extended surveillance or rescue missions. Optimizing power consumption and exploring alternative energy sources could mitigate this limitation.

Environmental Conditions: Extreme weather conditions, rough terrain, and unpredictable environments may pose challenges to the rover's functionality and reliability, requiring robust design considerations and adaptations.

Considering the limitations and searching for space of upgrading the project, we have some future works highlighted below:

Autonomous Navigation: Implementing advanced algorithms for autonomous navigation could enhance the rover's ability to navigate complex environments and make informed decisions based on sensor data, reducing reliance on manual control.

Sensor Fusion: Integrating additional sensors, such as GPS for location tracking and obstacle detection systems, could enhance the rover's environmental awareness and decision-making capabilities, improving its effectiveness in surveillance and rescue operations.

Machine Learning and AI: Leveraging machine learning and artificial intelligence techniques for object detection, anomaly detection, and predictive analytics could enable the rover to identify and respond to emergent situations more effectively, enhancing its utility in dynamic environments.

Addressing these limitations and pursuing future works can further enhance the capabilities, reliability, and effectiveness of the Surveillance Rover with Rescue Assistance Feature, making it a valuable asset in surveillance, emergency response, and disaster management scenarios.

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APPENDICES

Appendix A1 (Snapshots)

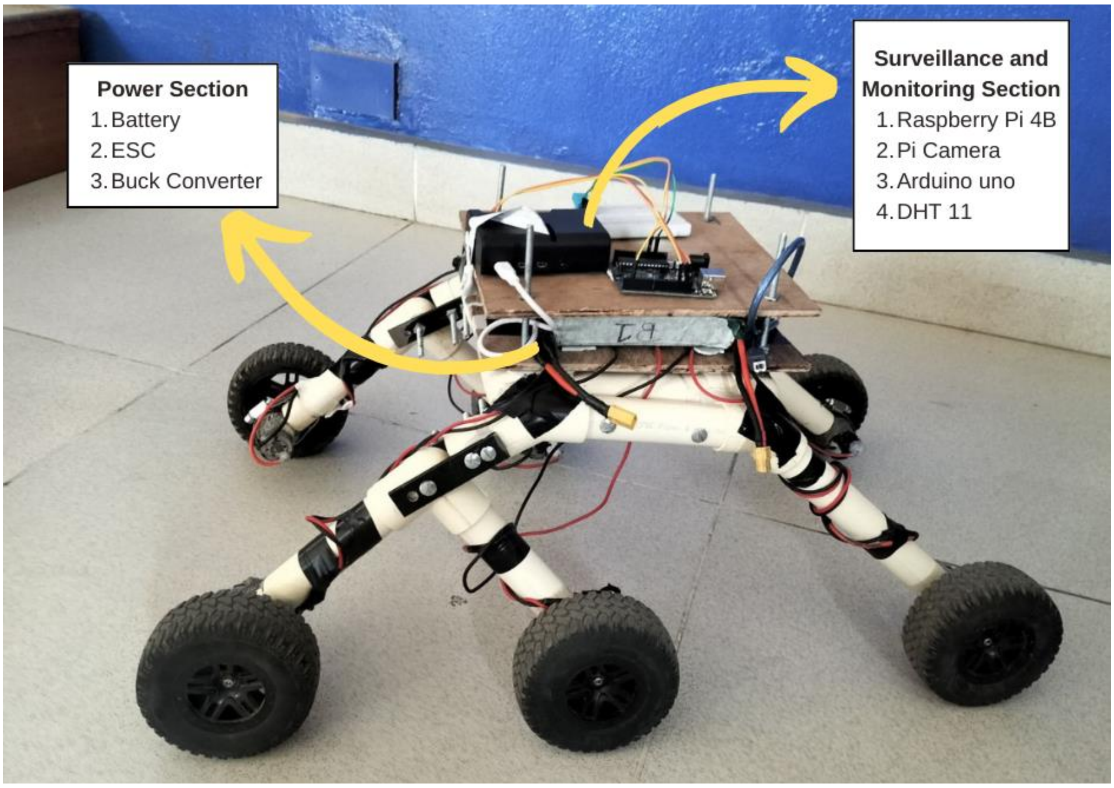

Figure A1. 1: Final Rover Design showing Component Sections

Figure A1. 2:Rover Testing on Basketball Court

Figure A1. 3:Rover Testing in the underground

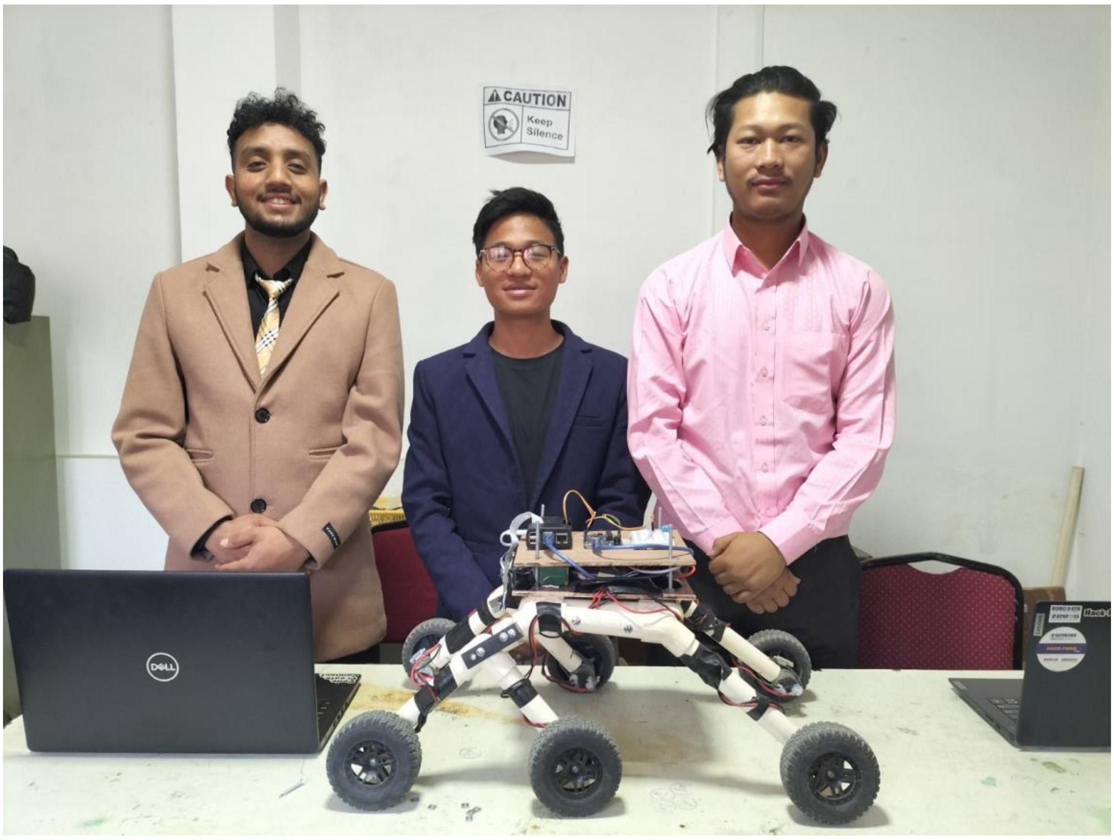

Figure A1. 4: Project team members

Appendix A2

1. Schedule

S.N.	ACTIVITY	AUG-SEP 2023	OCT-NOV 2023	DEC 2023	JAN-FEB 2024	MAR 2024
1.	Feasibility Study					
2.	System Specification					
3.	Requirement Analysis					
4.	Design • Design of Rover chassis					
	• Hardware interface					
	• Entire Hardware Assembling					
5.	Documentation					
6.	Testing					

Figure A2. 1: Gantt Chart

2. Cost Estimation

Table A2. 1: Cost Estimation

S.N.	Components	Details	Quantity	Price (Rs)
1.	Humidity and Temperature sensor	DHT-11	1	450
2.	LiPo Battery	3300 mAh	1	2500
3.	Pi camera	-	1	1900
4.	Motor Driver (ESC)	-	3	900
5.	Arduino uno	-	1	1000
6.	Transmitter Receiver	-	1	7000
7.	Buck Converter	-	1	300
8.	Raspberry Pi	Model 4 B	1	15000
9.	Rover Chassis	-	1	500
Total				29,550